Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Computer Vision, Deep Learning, Real-Time Edge Inference | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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14th October 2026

Re: AI Vision Engineer – Nexxa.AI

Dear Hiring Manager,

I am applying for the AI Vision Engineer role at Nexxa.AI. I am a machine learning engineer with a PhD from the University of Trento and 10+ years across industry and academia. For five and a half years at MAS Holdings I led production computer vision for manufacturing, from care-label inspection to loom-side fabric defect detection, and that work is the closest match to what you describe: vision systems for real industrial settings, taken from prototype to deployment.

The care-label QC system is a good example. I built it with OpenCV (template and feature matching, OCR, an explainable defect overlay) and iterated it three times, from a flatbed scanner to an industrial camera to a conveyor with a PLC reject. It cut inspection time by 99.5% and improved throughput by up to 300%. On the edge side, my published work at Trento runs real-time pattern detection in 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline, and I designed the benchmark suites and ablations that decided the architecture. I have also built the data and deployment pipelines around these models, using Python, PyTorch, Docker and AWS.

I should be direct about the gaps. My multimodal work is on audio, sensor and time-series data rather than image and text, so I have not yet fine-tuned vision-language models, and I have not worked with YOLO or Detectron2, or with Jetson, TensorRT and quantization in production. I have used HuggingFace Transformers and generative models (diffusion and VAE prototypes at McGill), and I expect to close these gaps quickly, since the evaluation, optimization and deployment fundamentals are the same.

Nexxa's focus on heavy industry is where I have done my best work, and I would like to bring that experience to your vision stack. I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, available immediately and comfortable working remotely. My portfolio, publications and open datasets are at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD